Food Rescue Optimization System

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***Abstract*— Food waste and food insecurity are two problems that still coexist in plenty of urban areas. A big part of the edible food is thrown away by restaurants, catering companies, and event organizers, whereas many families and community organizations are barely managing to get enough food for their needs. This article introduces FeedMePlz, a surplus food redistribution web platform that connects food donors with NGOs and community groups as verified receivers. The main feature of our platform is feeding. It allows donors to create a posting with information about the type of food quantity pickup location, and time availability while receivers can select, request, and make a final decision on suitable donations in real time. For better coordination, our platform has implemented role-based access control, location-aware matching, status tracking, and notification support. The purpose of the system is to prevent the wasting of food, to speed up the process of redistribution, and to make the donation process more transparent. After digitizing and streamlining food rescue operations, FeedMePlz is an example of how a simple web platform can be a support for social good, operational efficiency, and sustainable management of resources.**

Keywords:
Managing surplus food, reducing food waste, redistributing food, matching in real time, using location-based services, system of notifications, web-based platform, sustainable development, hunger mitigation, resource optimization, community support

Introduction

Food waste and hunger are a huge contradiction in today's society. Daily, working restaurants, event halls, hostels, and people's homes throw away edible food while at the same time, food shortages are experienced by less fortunate communities. Most of the time, it is not the production of food that is the problem, but a lack of a proper way to redistribute surplus food before it goes bad. Traditional donation methods are mostly informal, manual, and rely on personal networks. Since food is perishable, any delays in communication or transportation can quickly render a potential donation unusable. Consequently, food that could have been support to the communities nearby is very often wasted due to the lack of proper coordination, visibility, and accountability. Digital platforms offer a chance to make this process better. A web-based redistribution system can record donation details immediately, find a match between a donor and a recipient in the same area, and help in tracking the status from picking up to delivery. This will help in cutting down response time, building trust among the participants, and keeping a reliable record of transactions.

In this paper, we introduce FeedMePlz, a system for surplus food redistribution and tracking that aims to link donors and receivers via a single coordinated platform. Our solution focuses on making the system easy-to-use, providing real-time updates, ensuring transparency, and enabling location-aware interaction. The main objective is to change the way food donation is viewed - from a random emergency action that is reactive to a well-organized and reliable service process.

II RELATED WORK

Earlier studies on food waste indicate that a large percentage of human-edible food is wasted in the supply chains, households, and service environments. Researchers have been raising the alarm about how food waste is not just an economic inefficiency but also a social and environmental problem. As a result, redistribution of surplus food has come to the forefront as a key strategy for simultaneously reducing waste and enhancing food accessibility. Digital coordination of food rescue operations is one of the key points that studies and community-driven initiatives have highlighted as the sources of value. Web and mobile platforms can assist donors in disclosing food availability, recipients in discovering food opportunities, and drastically cutting the time for pickup and delivery. Some of the solutions focus on the importance of transparency, traceability, and more efficient resource allocation through the exchange of information at the right time. However, a large number of practical food rescue implementations tend to be quite narrow. For instance, many of the existing initiatives rely on semi-manual coordination, fragmented communication, or geographically enclosed operations. Such limitations can hinder scalability and result in delays, repetitive work, and unfavorable visibility into the status of each donation.

Building upon these prior works, the new FeedMePlz system integrates structured user roles, real-time listing and request processing, location-based coordination, and status notifications all in one web platform. Instead of concentrating on a single phase of the workflow only, the system is intended to help the whole donation process in a combined way.

III PROPOSED MODEL

The proposed model breaks down the food redistribution process into four main modules: user management, donation administration, matching and handling of requests, and tracking with notifications. These modules collectively facilitate a smooth and complete operational workflow for rescuing surplus food.

A. User Management and Authentication

The system recognizes donors, receivers, and administrators as separate user roles. Each person creates an account through a secure login process, and then the permissions associated with their role will limit what features they can access. Organizing user roles in this way enhances security, limits incorrect usage, and keeps the system manageable as the number of users increases.

B. Food Donation Management

Donors are able to list their donations by giving details about the kind of food, the estimated amount, how long it will be available, expiry or freshness limitations, and the location for picking up the food. Such data is entered into a central database and the relevant receivers are instantly able to view the donation. Donors are also capable of modifying or removing their donation listings, so the information is always up-to-date and correct.

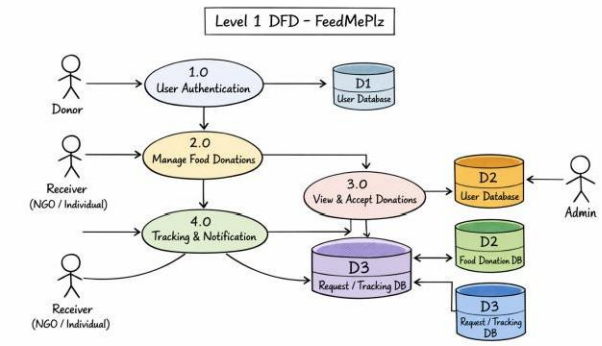
C. Matching and Request Handling

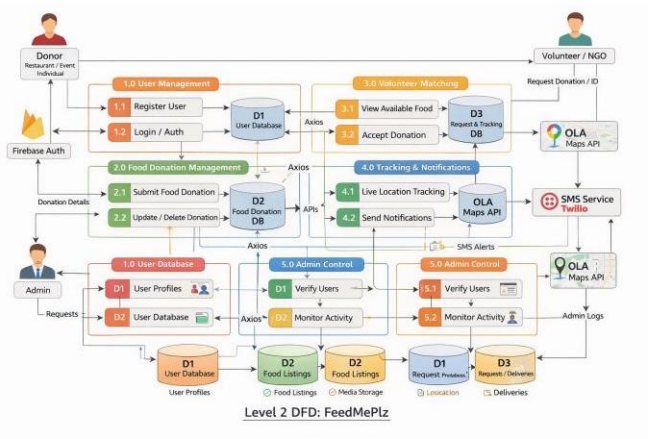
Recipients have a way to see what is available for donation aligned with their needs and locations. If a recipient comes across a donation that fits them, they can reach out via the platform. When a donation gets the green light, the system not only changes the state of the listing but also logs the exchange for both sides. In this way, the whole operation of finding and providing donations is streamlined, misunderstandings are almost eliminated, and the number of double claims are also minimized, plus the redistribution of items becomes accelerated.

D. Tracking and Notification Support

To make things more see-through, the platform features status changes being shared on different important phases like listing a new item, an item being taken, the item being picked up, and the item being delivered person to person, etc. Notifications play a crucial role in keeping the donor and the recipient on the same page. Additionally, location-based assistance for users to coordinate the fast collection of a donation. These kinds of tools become even more critical when it comes to perishable goods as their delay can also mean their unusable.

Module	Primary Function	Expected Benefit
User Management	Registers users and controls role-based access	Improves security and organized participation
Donation Management	Publishes and updates surplus food listings	Keeps donation data accurate and timely
Matching	Connects suitable receivers with listed donations	Reduces delay and duplicate claims
Tracking and Notifications	Monitors status and informs stakeholders	Improves transparency and coordination





IV CONCLUSION

FeedMePlz offers a very hands-on way of handling two very closely related issues - food waste that could have been avoided and food that people cannot get to easily. By paving the way for food donors and receivers to interact through a web interface, the system becomes a lot more effective at displaying surplus food and also reducing the time for redistribution. Instead of manual coordination being scattered, the platform brings in a more organized workflow that allows managing listings, handling requests, and keeping track of statuses. This increases accountability and helps with using surplus food before it turns bad. Besides making the operation run smoothly, the system being put forward has a hand in even bigger social and environmental things. A redistribution platform that people can count on can lead to community groups getting help, reducing the amount of work of waste stream and also pushing the use of resources in a more responsible way. So, in a way, FeedMePlz is not only a technical solution but also a highly socially relevant use of web technologies.

FUTURE SCOPE

The system could be improved by incorporating predictive analytics that forecast locations with a surge in demand, periods when donations peak, and potential areas of shortage. Armed with such data, the platform could transition from merely reacting to events to actually planning them. Besides this, the later editions of the software may connect directly with restaurant management systems, event organizers, or inventory control so that surplus food can be identified and advertised with minimal manual input. This, on the one hand, will encourage more users while, on the other hand, it will alleviate data entry responsibilities for the donors.

Benefits of route optimization, enhanced logistics support, and automated communication via SMS, email, or in-app messaging in increasing efficiency will not be missed. Including a feedback system for both donors and recipients will not only build up trust but also raise the standard of service. At a broader scale, the platform could be made accessible to multiple cities and be linked to networks of NGOs or public services. Impact reporting through live dashboards for instance meals served or food saved would significantly enhance transparency and help organizations in measuring their results in a more efficient manner.

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